

# CHAPTER 10: CELL CYCLE AND CELL DIVISION

Class 11 Biology • Unit 3: Cell Structure and Functions • Complete Comprehensive Guide

## Table of Contents

|                                        |          |
|----------------------------------------|----------|
| 10.1 Cell Cycle & Its Phases           | Page 1   |
| 10.2 M Phase (Mitosis & Cytokinesis)   | Page 2   |
| 10.3 Significance of Mitosis           | Page 3   |
| 10.4 Meiosis (Meiosis I & II)          | Page 3-4 |
| 10.5 Significance of Meiosis           | Page 5   |
| ★ NCERT Complete Solutions & Exercises | Page 5-6 |

## Core Concepts & Learning Objectives

All living organisms start life from a single cell. Growth and reproduction in cells occur through cyclic duplication of genome, synthesis of cellular components, and subsequent cell division. This document details the precise genetic and cytoplasmic events occurring during **Interphase**, **Mitosis**, and **Meiosis**.



Human Cell: 24h Span



Yeast Cell: 90 mins



Interphase: >95% Duration

## CHAPTER MIND MAP — CELL CYCLE AND CELL DIVISION

### 1. CELL CYCLE PHASES

#### A. Interphase (Prep Phase)

Cell grows & replicates DNA (>95% time)

- **G<sub>1</sub> Phase:** Growth, metabolic activity
- **S Phase:** DNA synthesis (2C→4C), Centriole duplicates
- **G<sub>2</sub> Phase:** Protein synthesis for mitosis

#### B. G<sub>0</sub> Phase (Quiescent)

Inactive stage, metabolically active (e.g. Heart cells)

### 2. MITOSIS (M PHASE)

#### Equational Division ( $2n \rightarrow 2n$ )

##### Prophase:

Chromosomes condense

**Metaphase:** Equatorial plate

**Anaphase:** Centromere splits

**Telophase:** Envelope reforms

**Cytokinesis:** Furrowing (Animals) vs Cell Plate (Plants)

### 3. MEIOSIS (REDUCTION)

#### Reduction Division ( $2n \rightarrow 4 \times n$ )

**Meiosis I:** Homologous chromosomes separate

##### Prophase I Sub-stages:

Leptotene → Zygotene (Synapsis) → Pachytene (Crossing over) → Diplotene (Chiasmata) → Diakinesis

**Meiosis II:** Sister chromatids separate (mitosis-like)

## 10.1 CELL CYCLE & PHASES OF CELL CYCLE

NCERT PAGES 120-122

### Definition: Cell Cycle

The sequence of events by which a cell **duplicates its genome**, synthesises the other constituents of the cell, and eventually divides into two daughter cells is termed the **Cell Cycle**.

*Note:* Cytoplasmic growth is a continuous process, whereas DNA synthesis occurs only during one specific phase (S phase).

### Two Basic Phases of Cell Cycle

#### 1. INTERPHASE ("Resting Phase" / Preparatory Phase)

> 95% of Duration

The period between two successive M phases where the cell prepares for division by undergoing both cell growth and DNA replication in an orderly manner.

##### G<sub>1</sub> Phase (Gap 1)

Interval between mitosis & initiation of DNA replication. Cell is **metabolically active & continuously grows**, but does not replicate DNA.

##### S Phase (Synthesis)

DNA replication occurs in nucleus. DNA content **doubles from 2C to 4C**. Chromosome number remains same (2n). Centrioles duplicate in cytoplasm.

##### G<sub>2</sub> Phase (Gap 2)

Proteins required for mitosis are synthesised while cell growth continues.

#### Quiescent Stage (G<sub>0</sub> Phase)

Cells in adult animals that do not exhibit regular division (e.g., **heart cells**) or divide only occasionally to replace lost cells exit G<sub>1</sub> phase to enter an inactive stage called **G<sub>0</sub> phase**. Cells remain **metabolically active** but no longer proliferate unless called upon to do so.

#### Mitosis in Haploid vs Diploid Cells

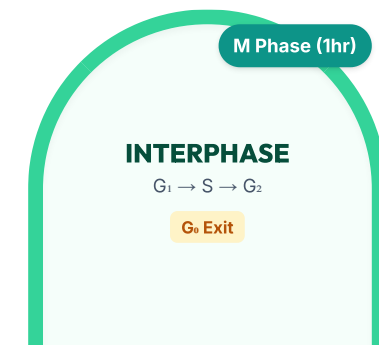
- Animals:** Mitotic cell division is restricted to **diploid somatic cells**. *Exception:* Male honey bees (drones) develop by haploid mitosis.

### Genetic Content Changes across Phases

NCERT Important Numerical Concept

| Phase                | Chromosome No.      | DNA Content         | Centriole Count |
|----------------------|---------------------|---------------------|-----------------|
| G <sub>1</sub> Phase | 2n (e.g., 16)       | 2C                  | 1 pair (2)      |
| S Phase              | 2n (Unchanged)      | 4C (Doubled)        | 2 pairs (4)     |
| G <sub>2</sub> Phase | 2n (16)             | 4C                  | 2 pairs (4)     |
| After M Phase        | 2n in each daughter | 2C in each daughter | 1 pair per cell |

### Diagrammatic View of Cell Cycle (Figure 10.1)



- **Plants:** Mitosis occurs in **both haploid and diploid cells** (e.g., gametophyte haploid plant body & sporophyte diploid body).

Human cells in culture divide once every ~24 hours. Interphase takes >23 hours.

### ★ ★ Exam Point: Onion Root Tip Example

Onion root tip somatic cell has **16 chromosomes** ( $2n=16$ ). At  $G_1$  it has 16 chromosomes & 2C DNA. After S phase it has **16 chromosomes & 4C DNA**. After M phase each daughter cell gets **16 chromosomes & 2C DNA**.

## 10.2 M PHASE (MITOSIS — EQUATIONAL DIVISION)

NCERT PAGES 122-124

Mitosis involves major reorganization of cellular components and preserves chromosome number, hence called **Equational Division**. It is divided into 4 progressive stages of karyokinesis followed by cytokinesis.

### 1. PROPHASE

First Stage

- Initiation of **chromatin condensation** to form compact mitotic chromosomes.
- Chromosomes consist of **two chromatids** attached at centromere.
- Duplicated centrosomes move towards opposite poles.
- Centrosomes radiate microtubules called **asters**. Asters + spindle fibres = **Mitotic Apparatus**.

✗ **Disappears:** Golgi complex, ER, nucleolus, & nuclear envelope.

### 2. METAPHASE

Morphology Stage

- Nuclear envelope completely disintegrates; chromosomes spread in cytoplasm.
- Chromosomes reach maximum condensation — **morphology best studied here**.
- Disc-shaped **kinetochores** on centromere serve as attachment sites for spindle fibres.
- All chromosomes align at the equator: **Metaphase Plate**.

🌀 **Key Feature:** Spindle attaches to kinetochores from opposite poles.

### 3. ANAPHASE

Splitting Stage

- Centromere of each chromosome **splits simultaneously**.
- Sister chromatids separate, now called **daughter chromosomes**.
- Chromatids migrate to opposite poles.
- Centromere leads towards pole at leading edge; arms trail behind.

⚡ **Key Feature:** Centromeres split; chromatids move to opposite poles.

### 4. TELOPHASE

Final Stage

- Chromosomes reach poles, **decondense & lose individual identity**.
- Chromatin material collects at opposite poles.
- Nuclear envelope develops around chromosome clusters at each pole.

✦ **Reappears:** Nucleolus, Golgi complex, & ER reform.

### 🔗 10.2.5 Cytokinesis: Animal Cell vs Plant Cell

### 🌀 Syncytium Formation

| Feature                           | Animal Cell Cytokinesis                  | Plant Cell Cytokinesis                             |
|-----------------------------------|------------------------------------------|----------------------------------------------------|
| <b>Mechanism</b>                  | Cell Furrow formation in plasma membrane | Cell Plate formation from center outwards          |
| <b>Direction</b>                  | <b>Centripetal</b> (outside to inside)   | <b>Centrifugal</b> (inside to outside)             |
| <b>Precursor / Middle Lamella</b> | Not required                             | Cell plate represents future <b>middle lamella</b> |
| <b>Cell Wall Restriction</b>      | Absent (Flexible membrane)               | Required due to inextensible rigid cell wall       |

In some organisms, **karyokinesis is not followed by cytokinesis**, resulting in a multinucleate condition known as a **syncytium**.

💡 **NCERT Example: Liquid endosperm in coconut.**

*Organelles like mitochondria & plastids get distributed between daughter cells during cytokinesis.*

### ✓ 10.3 Significance of Mitosis

- **Identical Genome:** Produces diploid daughter cells with identical genetic complement.
- **Multicellular Growth:** Growth of organisms occurs due to continuous mitotic division.
- **Nucleo-Cytoplasmic Ratio:** Restores nucleo-cytoplasmic ratio disturbed by cell growth.
- **Cell Repair & Replacement:** Continuously replaces cells of upper epidermis, gut lining, & blood cells.
- **Continuous Plant Growth:** Mitosis in apical & lateral cambium (meristems) allows lifelong plant growth.

### ✂ 10.4 Meiosis (Reduction Division) Key Features

**Cycles:** Two sequential nuclear/cell divisions (Meiosis I & II), but only **ONE single cycle of DNA replication** (during S phase before Meiosis I).

**End Result:** **Four haploid cells** are produced from one diploid parent cell.

**Pairing & Recombination:** Involves pairing of homologous chromosomes and crossing over between non-sister chromatids.

**Biological Function:** Ensures production of haploid phase in life cycle; fertilization restores diploidy.

### 🧬 10.4.1 MEIOSIS I — PROPHASE I (Longer & More Complex)

5 Sub-stages

#### 1. LEPTOTENE

Chromosomes become gradually visible under light microscope.  
Compaction of chromosomes continues throughout leptotene.

#### 2. ZYGOTENE

Chromosomes pair together:  
**Synapsis.**  
Paired chromosomes = **Homologous chromosomes.**  
**Synaptonemal Complex** forms.

#### 3. PACHYTENE

Tetrads become clearly distinct.  
**Recombination nodules** appear.  
**Crossing Over:** Exchange of genetic material between non-

Crucial

#### 4. DILOTENE

Dissolution of synaptonemal complex.  
Homologous chromosomes separate except at crossover sites.

#### 5. DIAKINESIS

**Terminalisation of chiasmata.**  
Chromosomes fully condensed.  
Meiotic spindle assembled.  
Nucleolus & nuclear envelope break down. Transition to

Complex called **Bivalent** or **Tetrad**.

sister chromatids of homologous chromosomes.  
Enzyme involved: **Recombinase**.

X-shaped structures formed:  
**Chiasmata**.  
*Note:* In vertebrate oocytes, can last months/years.

Metaphase I.

💡 🗨 **Easy Memory Trick for Prophase I Sequence:** "Lazy Zebras Pack Double Dice" → Leptotene → Zygotene → Pachytene → Diplotene → Diakinesis

### Metaphase I

Bivalent chromosomes align on equatorial plate. Microtubules attach to kinetochore of homologous chromosomes.

### Anaphase I

Homologous chromosomes separate to opposite poles. **Sister chromatids remain associated at centromeres**.

### Telophase I

Nuclear membrane & nucleolus reappear.  
Cytokinesis follows, producing a **Dyad of cells**.

### Interkinesis Stage

Stage between Meiosis I & II. Short-lived. **NO DNA replication occurs during interkinesis**.

## 10.4.2 & 10.5 MEIOSIS II & SIGNIFICANCE OF MEIOSIS

📖 NCERT PAGES 127-128

### Meiosis II (Resembles Normal Mitosis)

#### Prophase II

Initiated immediately after cytokinesis.  
Nuclear membrane disappears;  
chromosomes become compact again.

#### Metaphase II

Chromosomes align at equator. Microtubules attach to kinetochores of sister chromatids.

#### Anaphase II

Simultaneous splitting of centromeres. Sister chromatids move toward opposite poles.

#### Telophase II

Nuclear envelope reforms. Cytokinesis yields a **Tetrad of cells (4 haploid daughter cells)**.

### 🌞 10.5 Significance of Meiosis

1. **Conservation of Chromosome Number:** Conserves specific chromosome number of each species across generations in sexually reproducing organisms (paradoxically reduces number by half during gametogenesis).
2. **Genetic Variability:** Increases genetic variations in populations from one generation to next via crossing over. Variations are fundamental for **Evolution**.

| Basis of Comparison                     | Mitosis (Equational Division)                       | Meiosis (Reduction Division)                                                                           |
|-----------------------------------------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Cell Type</b>                        | Occurs in somatic cells (and germ cells for growth) | Occurs in specialized diploid germ cells during gametogenesis                                          |
| <b>Number of Divisions</b>              | Single nuclear and cell division                    | Two sequential nuclear and cell divisions (Meiosis I & II)                                             |
| <b>Daughter Cell Count &amp; Ploidy</b> | 2 diploid (2n) identical daughter cells             | 4 haploid (n) genetically varied daughter cells                                                        |
| <b>Synapsis &amp; Crossing Over</b>     | Absent; no pairing or recombination                 | Present in Prophase I (Zygotene & Pachytene)                                                           |
| <b>Anaphase Event</b>                   | Centromeres split; sister chromatids separate       | <b>Anaphase I:</b> Homologous chromosomes separate.<br><b>Anaphase II:</b> Sister chromatids separate. |
| <b>Biological Role</b>                  | Body growth, cell repair, tissue replacement        | Gamete formation, chromosome conservation, evolution                                                   |

## EXERCISES COMPLETE NCERT QUESTION-ANSWER SOLUTIONS

✓ QUESTIONS 1 - 16 ANSWERED

### Q1. Average cell cycle span for mammalian cell?

**Ans:** Approximately **24 hours** (Interphase ~23 hrs, M phase ~1 hr).

### Q2. Distinguish cytokinesis from karyokinesis.

**Ans:** *Karyokinesis* is division of nucleus (separation of chromosomes). *Cytokinesis* is division of cytoplasm separating cell into two daughter cells.

### Q3. Events taking place during interphase.

**Ans:** (i) **G<sub>1</sub>:** Cell growth, metabolic activity. (ii) **S:** DNA replication (2C→4C), centriole duplication. (iii) **G<sub>2</sub>:** Protein synthesis for mitosis & continuous growth.

### Q4. What is G<sub>0</sub> (quiescent phase)?

### Q8. Cytokinesis in plant vs animal cells?

**Ans:** Animal cells form a cell furrow in plasma membrane (centripetal). Plant cells form a cell plate starting from center outwards (centrifugal) due to rigid cell wall.

### Q9. Equal vs Unequal meiotic daughter cells?

**Ans:** *Equal size:* Spermatogenesis (4 equal sperm cells). *Unequal size:* Oogenesis (1 large ovum + small polar bodies).

### Q10. Anaphase of mitosis vs Anaphase I of meiosis?

**Ans:** In *Anaphase of mitosis*, centromeres split & sister chromatids separate. In *Anaphase I*, homologous chromosomes separate while sister chromatids remain attached at centromeres.

**Ans:** Inactive stage entered by cells exiting  $G_1$  that do not divide further (e.g., heart cells). Cells remain metabolically active but proliferate only when required.

**Q5. Why is mitosis called equational division?**

**Ans:** Because chromosome number in daughter cells remains exactly identical to that in parent cell ( $2n \rightarrow 2n$ ).

**Q6. Name the stage of cell cycle where event occurs:**

- (i) Chromosomes moved to spindle equator → **Metaphase**
- (ii) Centromere splits & chromatids separate → **Anaphase**
- (iii) Pairing between homologous chromosomes → **Zygotene (Prophase I)**
- (iv) Crossing over between homologous chromosomes → **Pachytene (Prophase I)**

**Q7. Describe: (a) synapsis (b) bivalent (c) chiasmata.**

**Ans:** (a) *Synapsis*: Pairing of homologous chromosomes during Zygotene. (b) *Bivalent*: Pair of synapsed homologous chromosomes (tetrad). (c) *Chiasmata*: X-shaped sites of crossover visible in Diplotene.

**Q11-12. Significance of Meiosis & Mitosis?**

**Ans:** Meiosis conserves chromosome number across generations and generates genetic variations for evolution. Mitosis enables growth, cell repair, and maintains genetic identity.

**Q14-15. Can there be Mitosis without S phase DNA replication / DNA replication without division?**

**Ans:** (Q14) No, mitosis cannot occur without DNA replication as genome must duplicate. (Q15) Yes, DNA replication without cell division leads to polyploidy/endomitosis.

**Q16. Parameters N & C change analysis across cell cycle:**

**G<sub>1</sub>:**  $N=2n$ ,  $C=2C$  • **S:**  $N=2n$ ,  $C=4C$  • **G<sub>2</sub>:**  $N=2n$ ,  $C=4C$  • **Post-Mitosis:**  $N=2n$ ,  $C=2C$  • **Post-Meiosis I:**  $N=n$ ,  $C=2C$  • **Post-Meiosis II:**  $N=n$ ,  $C=C$ .